

Lab-4: Web API

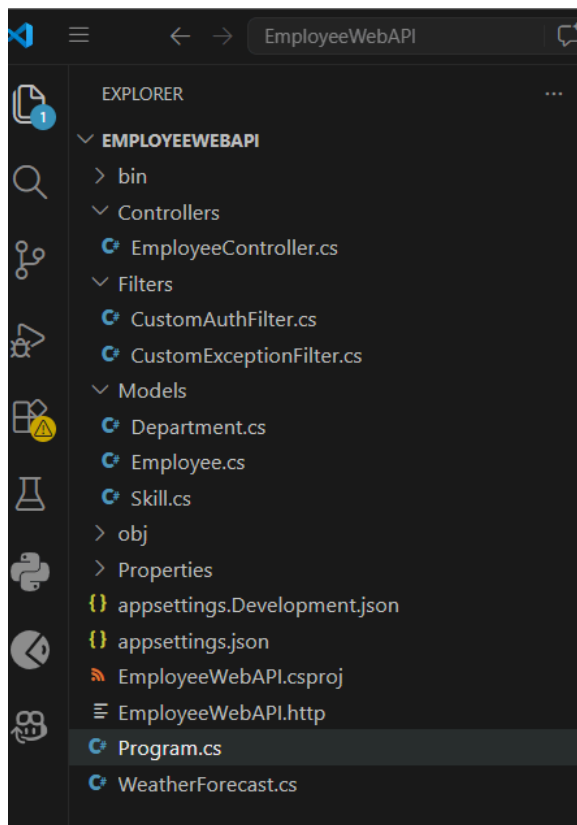
Aim

To implement CRUD operations in an ASP.NET Core Web API by performing update and delete operations on employee data using HTTP PUT and DELETE methods, and test the APIs using Swagger.

Objectives

- To create a Web API for employee data management.
- To implement the HTTP **PUT** method for updating employee details.
- To implement the HTTP **DELETE** method for deleting employee records.
- To use the **FromBody** attribute to receive data from the request body.
- To validate employee IDs and return appropriate error responses for invalid requests.

Folder Structure:



Program.cs:

```
1 using EmployeeWebAPI.Filters;
2 var builder = WebApplication.CreateBuilder(args);
3 // Add services to the container.
4 builder.Services.AddControllers(options =>
5 {
6 | options.Filters.Add<CustomExceptionFilter>();
7 | });
8 // Learn more about configuring OpenAPI at https://aka.ms/aspnetcore/openapi
9 builder.Services.AddOpenApi();
10 var app = builder.Build();
11 // Configure the HTTP request pipeline.
12 if (app.Environment.IsDevelopment())
13 {
14 | app.MapOpenApi();
15 | }
16 app.UseHttpsRedirection();
17 app.UseAuthorization();
18 app.MapControllers();
19 app.Run();
20
```

Output:

```
{
  "openapi": "3.0.1",
  "info": {
    "title": "EmployeeWebAPI | v1",
    "version": "1.0.0"
  },
  "servers": [
    {
      "url": "http://localhost:5256/"
    }
  ],
  "paths": {
    "/api/Employee": {
      "get": {
        "tags": [
          "Employee"
        ],
        "responses": {
          "200": {
            "description": "OK",
            "content": {
              "text/plain": {
                "schema": {
                  "type": "array",
                  "items": {
                    "$ref": "#/components/schemas/Employee"
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

```
PS C:\Users\suman\OneDrive\Desktop\EmployeeWebAPI> dotnet run
Building...
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: http://localhost:5256
info: Microsoft.Hosting.Lifetime[0]
      Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
      Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
      Content root path: C:\Users\suman\OneDrive\Desktop\EmployeeWebAPI
```

